

LA 6.2

$$B = \left(\begin{array}{c|c} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \end{array} \right) \quad \begin{matrix} k_1 \\ \downarrow \\ \end{matrix} \quad \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} = 1k_1 + 2k_2 \quad \begin{matrix} k_2 \\ \downarrow \\ \end{matrix} \quad \begin{pmatrix} 2 \\ 2 \\ -1 \\ -2 \end{pmatrix} = 2k_1 - 2k_2 \Rightarrow [A]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$|[A]_B - \lambda I| = \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2)$$

2: $\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \rightarrow \text{LO} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$

-3: $\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \rightarrow \text{LO} \left\{ \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right\}$

$$[A]_B^B = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}^{-1} = \overbrace{\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}}^P \overbrace{\begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}}^A \overbrace{\frac{1}{5} \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}}^{P^{-1}}$$

$$[A^n]_B^B = [A]_B^{B^n} = P A P^{-1} \cdot P A P^{-1} \dots = P \cdot A \cdot (P^{-1} P) A (P^{-1} P) A \dots P = P A^n P^{-1}$$

$$A^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) = [id]_{K_4}^B \cdot [A^n]_B^B \cdot \left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \frac{1}{5} \overbrace{\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}}^{\begin{pmatrix} 5 \\ 5 \end{pmatrix}} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2^n \\ (-3)^n \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 2 & -1 \\ 1 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 2^n \\ (-3)^n \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n \\ 2 \cdot 2^n - (-3)^n \\ 2^n + 2(-3)^n \\ 4 \cdot 2^n + 3(-3)^n \end{pmatrix}$$